

Muhammmad Athar Abbas
SP25-BSE-082
Object Oriented Programming
Assignment 3

Salat class (implementing Comparable and Comparator):

Program:

```
import java.util.Comparator;

public class Salat implements Comparable<Salat>{
    String name;
    int farz;
    int sunnah;
    Time time;

    public Salat(String name, int farz, int sunnah, Time time) {
        this.farz = farz;
        this.name = name;
        this.sunnah = sunnah;
        this.time = time;
    }

    @Override
    public int compareTo(Salat t) {
        if ("AM".equals(this.time.meridiem) &&
"PM".equals(t.time.meridiem)){
            return -1;
        }
    }
}
```

```

        if(this.time.meridiem.equals(t.time.meridiem)){
            if (this.time.hour != t.time.hour){
                return this.time.hour - t.time.hour;
            }
            if (this.time.minute != t.time.minute){
                return this.time.minute - t.time.minute;
            }
            if(this.time.second != t.time.second){
                return this.time.second - t.time.second;
            }
            return 0;
        }
        return 1;
    }

    @Override
    public String toString(){
        String message = name + " has " + farz + " farz" + ", " +
sunnah + " sunnah" + " and is held at: " + time.toString();
        return message;
    }

}

class sortByFarz implements Comparator<Salat>{
    @Override
    public int compare(Salat s1, Salat s2) {
        if(s1.farz != s2.farz){
            return s1.farz - s2.farz;
        }
        return (s1.name.compareTo(s2.name));
    }
}

```

```
    }  
}  
  
class sortBySunnah implements Comparator<Salat>{  
    @Override  
    public int compare(Salat s1, Salat s2) {  
        if(s1.sunnah != s2.sunnah){  
            return s1.sunnah - s2.sunnah;  
        }  
        return (s1.name.compareTo(s2.name));  
    }  
}  
  
class sortByName implements Comparator<Salat>{  
    @Override  
    public int compare(Salat s1, Salat s2){  
        return (s1.name.compareTo(s2.name));  
    }  
}
```

Time Class:

```
public class Time  
{  
    int hour;    // 0 - 12  
    int minute; // 0 - 59  
    int second; // 0 - 59  
    String meridiem; //AM - PM
```

```
public Time( int h, int m, int s, String meri )
{
    hour = h;
    minute = m;
    second = s;
    meridiem = meri;
}
// convert to String in standard-time format (H:MM:SS AM or
PM)
public String toString()
{
    return String.format( "%d:%02d:%02d %s",
        hour,minute, second, meridiem);
}
}
```

Driver Class:

```
import java.util.ArrayList;
import java.util.Collections;

public class DriverClass {
    public static void printList(ArrayList<Salat> salatList){
        for(Salat s : salatList){
            System.out.println(s);
        }
    }
    public static void main(String[] args) {
        Time fajarTime = new Time(6,0,0,"AM");
        Salat fajar = new Salat("Fajar",2,2,fajarTime);
    }
}
```

```
Time zuhurTime = new Time(1,30,0,"PM");
Salat zuhur = new Salat("Zuhur",4,6,zuhurTime);
Time asarTime = new Time(4,0,0,"PM");
Salat asar = new Salat("Asar",4,4,asarTime);
Time maghribTime = new Time(5, 12, 0, "PM");
Salat maghrib = new Salat("Maghrib",3,2,maghribTime);
Time ishaTime = new Time(7,0,0,"PM");
Salat isha = new Salat("Isha",4,6,ishaTime);

ArrayList<Salat> salatList = new ArrayList<>();

salatList.add(maghrib);
salatList.add(fajar);
salatList.add(zuhur);
salatList.add(isha);
salatList.add(asar);

System.out.println("Unordered: ");
printList(salatList);
//default Ordering
System.out.println("\nOrdered by Time: ");
Collections.sort(salatList);
printList(salatList);

System.out.println("\nOrdered by Name: ");
Collections.sort(salatList, new sortByName());
printList(salatList);

System.out.println("\nOrdered by Farz: ");
Collections.sort(salatList, new sortByFarz());
printList(salatList);
```

```
        System.out.println("\nOrdered by Sunnah");  
        Collections.sort(salatList, new sortBySunnah());  
        printList(salatList);  
  
    }  
}
```

(output on next page)

Output:

```
[athar@athar ~/semester2/OOP/java_programs/Theory_Assignments/3]$ java DriverClass
Unordered:
Maghrib has 3 farz, 2 sunnah and is held at: 5:12:00 PM
Fajar has 2 farz, 2 sunnah and is held at: 6:00:00 AM
Zuhur has 4 farz, 6 sunnah and is held at: 1:30:00 PM
Isha has 4 farz, 6 sunnah and is held at: 7:00:00 PM
Asar has 4 farz, 4 sunnah and is held at: 4:00:00 PM

Ordered by Time:
Fajar has 2 farz, 2 sunnah and is held at: 6:00:00 AM
Zuhur has 4 farz, 6 sunnah and is held at: 1:30:00 PM
Asar has 4 farz, 4 sunnah and is held at: 4:00:00 PM
Maghrib has 3 farz, 2 sunnah and is held at: 5:12:00 PM
Isha has 4 farz, 6 sunnah and is held at: 7:00:00 PM

Ordered by Name:
Asar has 4 farz, 4 sunnah and is held at: 4:00:00 PM
Fajar has 2 farz, 2 sunnah and is held at: 6:00:00 AM
Isha has 4 farz, 6 sunnah and is held at: 7:00:00 PM
Maghrib has 3 farz, 2 sunnah and is held at: 5:12:00 PM
Zuhur has 4 farz, 6 sunnah and is held at: 1:30:00 PM

Ordered by Farz:
Fajar has 2 farz, 2 sunnah and is held at: 6:00:00 AM
Maghrib has 3 farz, 2 sunnah and is held at: 5:12:00 PM
Asar has 4 farz, 4 sunnah and is held at: 4:00:00 PM
Isha has 4 farz, 6 sunnah and is held at: 7:00:00 PM
Zuhur has 4 farz, 6 sunnah and is held at: 1:30:00 PM

Ordered by Sunnah
Fajar has 2 farz, 2 sunnah and is held at: 6:00:00 AM
Maghrib has 3 farz, 2 sunnah and is held at: 5:12:00 PM
Asar has 4 farz, 4 sunnah and is held at: 4:00:00 PM
Isha has 4 farz, 6 sunnah and is held at: 7:00:00 PM
Zuhur has 4 farz, 6 sunnah and is held at: 1:30:00 PM
[athar@athar ~/semester2/OOP/java_programs/Theory_Assignments/3]$
```